

RUMING LIU

Hoboken, NJ | Ph.D. candidate, CPT in 2026; expected graduation May 2027

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Education

Stevens Institute of Technology

Doctor of Philosophy, Finance (Researches: Cryptocurrency, Blockchain & Quantitative Finance)

Sep. 2022 – May. 2027

Hoboken, NJ, U.S.A

University of Southern California

Master of Science, Mathematical Finance

Sep. 2019 – Jun. 2021

Los Angeles, CA, U.S.A

Jiangxi University of Finance and Economics

Bachelor of Economics, Finance

Sep. 2015 – Jun. 2019

Nanchang, Jiangxi, China

University of California San Diego

Extension Program, Statistics

Mar. 2018 – Jun. 2018

La Jolla, CA, U.S.A

Work Experience

Moody's Analytics

Financial Engineer Specialist

Sep. 2021 – Sep. 2022

San Francisco, CA, U.S.A

- Provided thorough quantitative and conceptual solutions for default risk to clients.
- Supported on products [RiskCalc](#), [CreditEdge](#), [RiskFrontier](#) and [MIS rating](#) via web, application, and API.
- Worked on Moody's credit model (Probability of Default, Expected Default Frequency, Loss Given Default, Distance to Default, Implied Rating, G-Correlation, Credit Migration, Risk Contribution, etc.)

China Investment Securities

Derivatives Analyst

Dec. 2018 – Mar. 2019

Shenzhen, Canton, China

- Designed, built and maintained tools used in dynamic hedge program and derivative portfolio management.
- Assisted in developing option price forecasting model.
- Developed a detailed knowledge of volatility arbitrage and market maker trading strategy.
- Provided Front Office support for financial reporting and data mining.

China Merchants Bank

Financial Data Analyst

Dec. 2017 – Jan. 2018

Shenzhen, Canton, China

- Took part in Bitcoin market arbitrage analysis research, worked in trading strategy and algorithm part.
- Worked with other programmer in financial products classification project by machine learning technique.
- Involved in design, testing and implementation of new financial and managerial reports.
- Wrote and published regular and ad hoc macro-economy reports.

Selected Researches

Flow-Induced Momentum in Mutual Funds: Evidence from Crypto Exposure ([Link](#))

Apr. 2026

Attention Based Large Factor Model ([Link](#))

Mar. 2026

The Impact of Spoofing on Bitcoin Market Microstructure ([Link](#))

Mar. 2025

Price Dispersion under Arbitrage Frictions: Evidence from Centralized and Decentralized

Mar. 2024

Cryptocurrency Exchanges ([Link](#))

Sentiment in the Cross Section of Cryptocurrency Returns ([Link](#))

Dec. 2023

More on homepage ([Link](#))

Technical Skills

Experienced with Blockchain technology: Bitcoin, Ethereum and other blockchain protocols. Competent in Blockchain development (DApps) language Solidity, Brownie framework, Rust and Vyper. Good at using statistical tools such as Matlab, SPSS, SAS and R to conduct quantitative research. Familiar with statistical inference and machine learning by using Python. Skilled in API and web mining. Have experience in MySQL, MongoDB, PostgreSQL and data mining. Proficient in AI-assisted development tools including Claude Code and Codex. Experienced with LangGraph framework for building AI agent applications. Passed FRM examination Level I in 2017.

Last updated September 15, 2026.